

APM Project Management Qualification

Learner Study Pack



Risk and issue management

Introduction

This section of the Learner Study Pack will cover the Risk and Issue Management competence of the APM Project Management Qualification.

This section will enable you to gain an understanding of the benefits, purpose and role risk and issue management provides to support successful project delivery.

Learning objective and outcomes

Learning objective:

23. Understand risk and issue management as the ability to identify and monitor risks (threats and opportunities); plan responses to those risks and respond to issues.

By the end of this activity, you should be able to:

23a) Understand the benefits of risk and issue management and the role of contingency planning in projects.

23b) Understand the purpose and importance of each stage in a risk management process (such as identification, analysis, monitoring, escalation, response and closure) and an issue management process (such as logging and analysis, escalation, and assignment of actions)

And

Understand why these stages are different for linear and iterative life cycles.

23c) Know proactive and reactive responses to risks (such as avoid, reduce, transfer or accept and exploit, enhance, share and reject).

23d) Understand why governance is important in risk and issue management.

Study time

This section of the Learner Study Pack will take approximately 1 - 2 hours to complete.

Recommended resources

In addition to your Learner Study Pack, use the below references for more information on Risk and Issue Management:

- APM Body of Knowledge Seventh edition, 4.3.3 (Risk management); 4.2.2 (Risk identification); 4.2.3 (Risk analysis); 1.3.2 (Assurance principles)
- APM Learning Module, Information sheet: Risk management, and Information sheet: Issue management
Accessed at: <https://www.apm.org.uk/resources/apm-learning/>
- APM Resource, What is risk management?
Accessed at: <https://www.apm.org.uk/resources/what-is-project-management/what-is-risk-management/>
- APM Risk Specific Interest Group (2018), Project risk analysis and management
Accessed at: <https://www.apm.org.uk/resources/white-papers/project-risk-analysis-and-management/>
- APM Resource, Risk appetite and risk tolerance
Accessed at: <https://www.apm.org.uk/resources/find-a-resource/risk-appetite-and-risk-tolerance/>

Overview

Risk management involves identifying potential uncertainties, assessing their impact, and developing strategies to mitigate or capitalise on them.

Whereas issue management focuses on addressing problems as they arise during the project execution phase.

Together, these processes help project teams navigate uncertainties and challenges, ultimately enhancing the project's ability to stay on track and achieve its objectives.

Key Aspects of Risk Management

Risk management allows individual risk events and overall risk to be understood and managed proactively, optimising success by minimising threats and maximising opportunities.

Risk management is influenced by the risk appetite of the investing organisation that will provide the tolerance levels on what is acceptable or not.

The risk management process is iterative to reflect the dynamic nature of project-work.

Benefits

The benefits of risk management include:

- Supports more realistic plans, schedules and budgets.
- Increases the likelihood of schedules and budgets being kept to.
- Greater accuracy in allocated contingencies through more meaningful assessment.
- Supports continuous learning and improvement.
- All contributing to increasing stakeholder confidence.

Key Aspects of Issue Management

The process used to manage issues is a simple concept looking at first understanding the issue to prioritise and assign owners for a timely resolution.

Issue resolution requires support from the project sponsor who may escalate to the governance board.

The management of issues are tracked from identifying to resolving the issue and may require a re-plan of the baselined project management plan.

Benefits

- Provides the structure and escalation routes required to resolve issues in a timely manner.
- Raises visibility amongst senior stakeholders who have access to organisational resource to support resolution.
- Provides a consistent way to document issues so they can be assessed against the plan.
- Enables re-prioritisation and decision making.

- Ultimately it enables issues to be resolved efficiently so work can progress preventing further delays.
- Informs continuous learning and improvement.

Contingency Planning

Contingency is the resource set aside for responding to identified and unidentified risks also known as the 'unknown unknowns.'

Contingency is not hidden extra time or money to deliver planned scope, it is clearly identified and is there to cover the gap between the plan and known risks.

In addition to the contingency for known risks, some projects will have access to a management reserve for unidentified risks or identified risks that have very low likelihood of materialising but if they did would have a very high impact.

Contingency is most typically expressed in two ways:

1. Budget
2. Time.

It's for dealing with uncertain events that have an impact on either the project costs or financial benefits or the project timescales.

However, if using an iterative life cycle and timeboxes, the project would consider contingency in terms of scope or quality.

Where risks are identified, the planned contingency has provision to cover the gap, and the Project Manager requests to drawdown some of the contingency to manage mitigation.

Risk Management Process

Risk management must be initiated by first establishing the risk management plan for the project.

This plan provides the roles and responsibilities, how risks are to be assessed and the impact evaluated, agreed tolerances for risk thresholds, reporting and escalation points.

It also serves as an important reference for all stakeholders.

Sometimes Initiation is the first step, but a typical risk management process will then include the following steps:

1. Identification

- Identifies the risk events relevant to a project considering risk appetite.
- Ongoing process throughout the project life cycle.
- There are several tools and techniques that can be used such as, SWOT analysis, brainstorming, checklists, analysis of assumptions and constraints, and prompt lists, to draw out knowable risks.
- Once identified, risks should be documented with the appropriate information in the project risk register and an owner assigned to assess and manage.
- When working in an iterative life cycle, risk identification has a different focus, using a risk-based approach mindset to influence the design of each iteration, whilst continuing to look at what is risky, why and what can be done about it during deployment.

2. Analysis

- Assess the relative impact of the identified risk on the project objectives.
- Project manager will undertake this, with the support of the project team and subject matter experts using qualitative and or quantitative techniques.
- Qualitative analysis looks at the impact on objectives, assessing the probability / likelihood of occurrence and the size of impact on schedule, cost, benefits and other objectives, plotting the risks on a probability/impact grid.
- Quantitative analysis determines the combined effect of risks on objectives and for focus on specific risk-based decisions using techniques such as, Monte-Carlo simulation, decision trees and sensitivity analysis.
- When working in an iterative life cycle, risks will be analysed for each iteration, the release of each release could then lead to different risks to be analysed.

3. Monitoring and Escalation

- Monitoring of risks for occurrence and knowing when to escalate.
- If risk thresholds are reached and considered to have potentially critical consequences to the project objectives, escalation would be required.

4. Response

- Considers the appropriate response to put in place to deal with the risk, whether proactive or reactive.
- Determines whether the risk is deemed a threat or an opportunity.

- Determines whether it makes sense to proactively invest in mitigation activity to bring the risk back within tolerance levels.
- Secondary risks or residual risks may emerge because of the response and should be considered.
- Focus is on putting a risk efficient solution in place, increasing the likelihood of project success.

5. Closure

- Closes the risk in the risk register when it no longer poses a risk to the project. This could be because it has been removed, responded to, occurred, or passed the point it was likely to occur.
- As the last step in the process, it is important to update the Risk Register with information relating to the risk, closure and lessons learned to avoid unnecessary work of monitoring risks that will no longer occur and to support continuous improvement of project activities and processes.
- Any remaining risks at project closure should be communicated to stakeholders involved in the handover and adoption phase.

Issue Management Process

Tracking the management of issues happens from identification through to resolution, including any change control and replanning activities required to the baselined project management plan.

The Issue Management process includes the following steps:

1. Log and Analyse

- An issue register / log should be maintained and used by the project manager and team to record, analyse, prioritise and track the status until the issue is resolved.
- Prioritisation is based on the success criteria and benefits for the work, considering scope, quality, time, cost and benefits in the business case.
- Issue registers / logs are a source of knowledge for future projects.

2. Escalate

- The issue will need escalating to the project sponsor, who may then escalate to the governance board.
- The sponsor / governance board will help the project manager define the best course of action for a resolution.

3. Assign Actions

- Actions are assigned to a person or group best placed to resolve.
- Look to implement a resolution in a timely manner.

4. Apply Change Control

- Issues that result in a change to scope or the baselined plan will need to be taken through the change control process.
- This ensures all project outputs are updated and checked for no further impact to the project.

Proactive and Reactive Responses to Risks

Risks are categorised as threats or opportunities and have two main types of responses to them, either proactive or reactive:

Proactive

A planned and implemented response undertaken to address the likelihood of the risk occurring or the size of the impact if it did occur.

Actions taken can include:

Threats

- Avoid - change objectives or practices to change the cause of the threat so it can no longer occur.
- Reduce - take action to reduce the probability of the threat occurring or the impact on the project.
- Transfer - from one party to another passing on the responsibility for bearing the impact of the threat.

Opportunities

- Exploit - change objectives or practices to implement a response to maximise the opportunity to gain maximum benefit / advantage.
- Enhance - the probability of the opportunity eventually being exploited by increasing the likelihood and / or impact.
- Share - collaborating with other organisations to increase the chances of the opportunity being realised.
- Reject - decide not to take advantage of the situation as the opportunity is of little value or the work to gain the result is too great.

Reactive

Threats

A provision for a course of action that will only be implemented if the risk materialises.

Actions taken can include:

- Accept - no acceptable or viable approaches to take to avoid or reduce, the threat and any residual risk must be managed, contingency plans implemented should the risk materialise

Governance in Risk and Issue Management

Governance and risk management work together providing systems of control for the project, giving confidence to stakeholders.

It informs risk and issue management through the processes, procedures, standardised methods, specialist expertise and resources.

It sets out the governance arrangements for the organisation and project, the life cycle to be used for project delivery, the governance route for reporting and escalating, and provides assurance which is targeted where the greatest risks exist.

Assurance is objective and independent, and governance relies on assurance to support effective decision making when it comes to managing risks and issues.

Risk analysis provides guidance to assurance on where vulnerabilities lie and in return evidence from assurance improves risk assessment as well as informing decision making at gate ways.

Scenario One:

During the initial planning of the 'Snappy Appy' renovation, the architect shows you their ambitious and somewhat complex architectural designs.

Designs of this nature will require specialised construction techniques and you have been asked to identify any potential risks that may impact the project as part of creating an effective and practical risk and issue management process.

What are the key benefits of creating a comprehensive risk management plan?

What tools or techniques could you use to help identify potential risks?

Why is governance important when creating a comprehensive risk management plan?

How could you identify potential mitigations for any potential risks?

Scenario Two:

One of the subcontractors working on the foundations of the new parts of the building has approached you with a concern around supplies and resources. They have been told by their current supplier that due to the growing price of steel, there could be a potential increase to the overall budget of £50,000.

They are worried that if they spend time trying to find an alternative supplier this could affect the tight timeline for the renovation project and are looking for some help to resolve the potential problem without delaying the build.

What steps from the Issue Management process could you follow to ensure minimal impact on the overall project?

Using the Issue Management Log below, complete the details for this issue:

<i>Issue No</i>	<i>Description</i>	<i>Priority</i>	<i>Category</i>	<i>Reported by</i>	<i>Assigned to</i>	<i>Status</i>	<i>Date to resolve</i>	<i>Notes</i>

Scenario Three:

As the property is an historical home, you have contracted a team of specialised Stonemasons to repair and replace the stone frames around the windows and doors of the building.

The Stonemasons previously worked on the office on a smaller job last summer, when they were 4 weeks late in starting than originally agreed. You are concerned that this may happen again and have logged this as a risk.

Using the table below, what actions could you take for each type of response to this risk?

Risk Response	Actions
Avoid	
Reduce	
Transfer	
Exploit	
Enhance	
Share	
Reject	

What risk response would you recommend selecting?

Learning review questions

Before you attempt the exam practice questions in the next section, we recommend you review your learning to quickly see which aspects you don't understand or can't remember as well.

It will also boost your confidence by confirming what you have learnt and your demonstrated ability to apply that knowledge, so that you are ready to practise for the exam.

Here are some questions you may wish to answer to review your learning from the Risk and Issue Management section:

- What is the benefit of having a Risk Management Plan?
- How might the Project Sponsor support issue resolution?
- How can assurance activities inform risk assessments?
- In what situations should a project manager drawdown contingency?
- Why is continuous monitoring crucial in risk management?
- What is the difference between quantitative and qualitative analysis?
- What are the potential constraints that can impact issue resolution?

Exam practice questions

23.1 At the outset of the project, you lead a risk analysis with your project team.

Which of these is not a benefit of risk analysis?

(1 mark)

- a. Gain a better understanding of the project's risk appetite.
- b. Discourages the acceptance of financially unviable solutions.
- c. Ensures project risks do not become issues.
- d. Allows a more meaningful assessment and justification of contingencies.

23.2 Which of the statements below most accurately reflects the correct order of stages in a risk management process?

(1 mark)

- a. Initiation, Identification, Analysis, Monitoring, Response, and Closure.
- b. Monitoring, Identification, Escalation, Response, and Closure..
- c. Initiation, Analysis, Escalation, Monitoring, Response, and Closure.
- d. Analysis, Monitoring, Escalation, Response and Closure.

23.3 You are managing a project to build a new development of 100 luxury homes.

During the analysis phase, it is identified that the location of the new development is regularly subjected to poor winter weather conditions which could delay the project's timescales and increase costs. You log this as a risk in the project's risk register and plan a risk response to 'transfer the risk.'

Which of these actions is the best example of how to transfer this risk?

(1 mark)

- a. Take out liability insurance to cover the costs of any delays due to poor weather.
- b. Move the development to a new location that has no risk of poor weather.
- c. Ask the project sponsor to sign off on the project despite the poor weather risk.
- d. Change the project scope to build 50 houses in winter, and 50 houses in summer.

23.4 You are in the early stages of a project and are responsible for drafting the risk management plan, including establishing a clear system of governance.

Which two of the following are not benefits of governance when thinking about risk management?

(1 mark)

1. Governance details when and how to escalate risks and clearly delegates authority.
2. Governance provides clear risk thresholds, in line with the organisation's risk tolerance.
3. Governance prevents the project manager from quickly making decisions about risks and how to respond to them.
4. Governance ensures that risk is managed consistently across a company portfolio to enable meaningful reporting and lessons learnt.
5. Governance prevents risk events from occurring through early detection.
6. Governance increases trust in the project's ability to identify and manage risk.

Select the correct combination of statements:

- a. 1 & 4
- b. 3 & 6
- c. 3 & 5
- d. 5 & 1

23.5 What are five reasons why risk management is important in projects? (5 marks)

23.6 You are managing a project to implement a new financial management system. During testing, the software crashes unexpectedly under specific user conditions.

The testing team choose not to log this bug as an issue.

What two impacts could choosing not to log this as an issue have on the project?
(2 marks)

23.7 Give one response to risk and describe what it involves.
(2 marks)

23.8 As a project progresses, the project team recognise that an identified risk has exceeded the tolerated threshold.

____(a)____ provides the project team with an overview of what is expected of them when managing risk throughout the duration of the project.

When a risk exceeds its tolerated threshold, the first step of a project team member should be informing the ____ (b) ____.

(2 marks)

(a)

1. Risk identification
2. The RAID Log
3. Risk appetite
4. Clear roles and responsibilities

(b)

1. Project sponsor
2. Project manager
3. Project team
4. Subject matter expert

Exam practice answers

Question	Answer / Mark Scheme
23.1	C
23.2	A
23.3	A
23.4	C
23.5	1 mark for each correct reason why risk management is important, 5 marks available in total. For example, but not limited to: • Supports more realistic / accurate plans / schedules. • Supports more realistic / accurate budgets. • More likely that schedules / budgets will be followed. • More accurate project / schedule / budget estimates can be made. • Helps engage stakeholders by showing that risks are being considered. • Helps build confidence that the project will be successful as risk is being managed. • More accurate requirements for any contingency can be made. • Supports continuous learning and improvement. • Allows the project manager to know the tolerance levels for each risk. • Contributes to project success as threats can be minimised. • Contributes to project success as opportunities can be maximised.
23.6	1 mark for each impact this could have on the project, 2 marks available in total – any of the below: • No official record of the project means other people /stakeholders / the project manager / other testers / developers are unaware of the problem. • If the issue isn't known, it cannot be resolved / risk it may never be fixed. • The bug could lead to other issues during future testing / when the system goes live. • The testing team cannot have support from the wider project team in fixing the bug. • Overall quality of the project is impacted by the bug not being fixed / issue not being resolved. • Miss the opportunity to learn from the bug / from the issue.
23.7	1 mark for naming a correct response to a risk – any of the below: • Avoid • Reduce / Mitigate • Transfer • Accept • Exploit • Enhance • Share • Reject 1 mark for describing a response to risk correctly – any of the below: • Avoid – change the circumstances that lead to the risk, so it can no longer happen. • Reduce / Mitigate – take action to reduce the probability of the risk occurring, or reduce the impact if it did happen. • Transfer – transfer risk to someone else / another party, so they would be impacted. • Accept – live with the risk, as no other response is viable / acceptable. • Exploit – decide to take advantage of the opportunity.

Question	Answer / Mark Scheme
	• Enhance - change the circumstances that lead to the risk, so it's more likely to happen / would have greater positive impact if it did happen. • Share – collaborate with someone else / another party, to take advantage of the opportunity. • Reject – decide not to take advantage of the opportunity due to low benefits or too much effort involved.
23.8	(a) 4 (b) 2